- Non-existence of the derivative of $y = f(x)$ at $x = x_0$ implies that the graph of $y = f(x)$ fails to admit tangent at $(x_0, f(x_0))$.
- Geometrically, if the graph of $y = f(x)$ admits cups $(\vee)$ or caps $(\wedge)$ at $x = x_0$ then derivative at $x = x_0$ does not exist.
- Derivative should be understood as a process not as a set of rules.
- Differentiability implies continuity, but the converse is not true.
- The sum, difference, product and composite of differentiable function is differentiable, and the quotient of two differentiable function is differentiable wherever it is defined.

(i) $\dfrac{d}{dx}(f(x) \pm g(x)) = \dfrac{d}{dx} f(x) \pm \dfrac{d}{dx} g(x)$

(ii) $\dfrac{d}{dx}(f(x)g(x)) = f(x)\dfrac{dg}{dx} + g(x)\dfrac{df}{dx}$

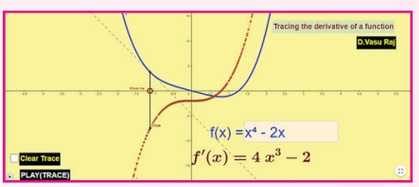
(iii) $\dfrac{d}{dx}((fog)(x)) = f'(g(x))g'(x)$

(iv) $\dfrac{d}{dx}\left(\dfrac{f(x)}{g(x)}\right) = \dfrac{g(x)f'(x) - f(x)g'(x)}{g^2(x)}$, where $g(x) \neq 0$.

Differential Calculus

ICT CORNER 10(a)

Expected Outcome



Step 1

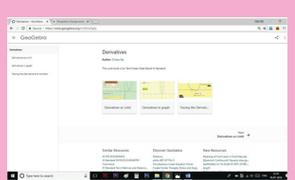
Open the Browser type the URL Link given below (or) Scan the QR Code.

GeoGebra Workbook called "Derivatives" will appear. In that there are several worksheets related to your lesson.

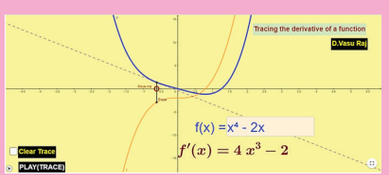
Step 2

Select the work sheet "Tracing the derivative of a function". You can enter any function in $f(x)$ box. You can see the function in blue colour and derivative in orange colour. Click play trace button to get animation of the locus of derivative(x, slope at x)

Observe the trace and find that derivative is the path of slope at each point on $f(x)$.



Step1



Step2

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Browse in the link:

Derivatives: https://ggbm.at/fk3w5g8y

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